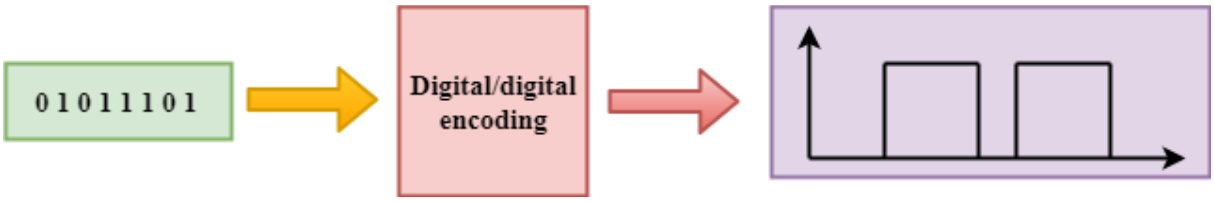
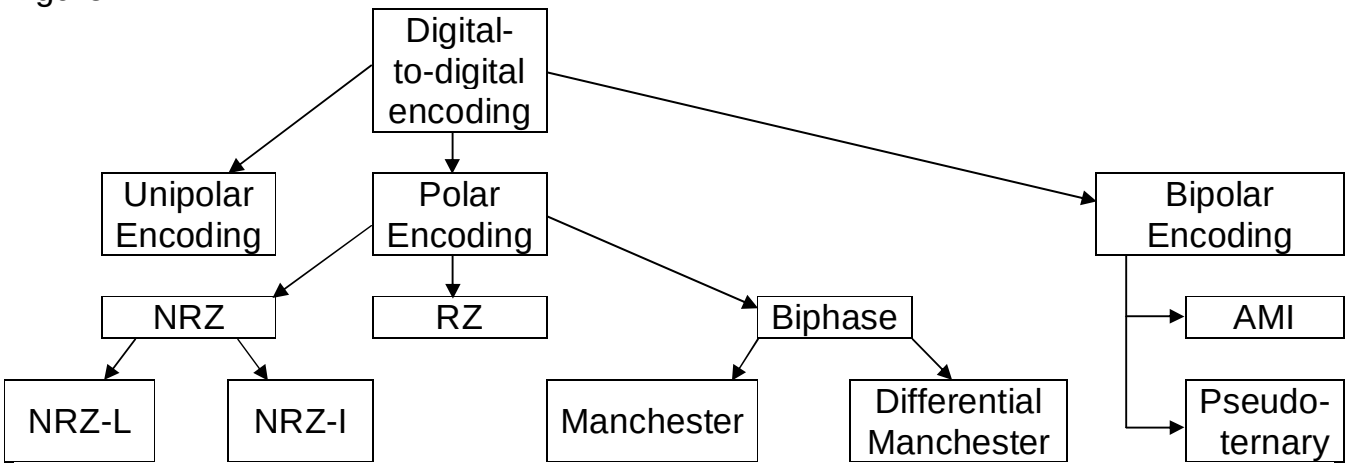
CORE – 9: Computer Networks (Unit–2)

Digital-to-Digital Conversion: Digital-to-digital encoding is the conversion of digital data to digital signal. When binary 1s and 0s generated by the computer are translated into a sequence of voltage pulses that can be propagated over a wire; this process is known as digital-to-digital encoding.

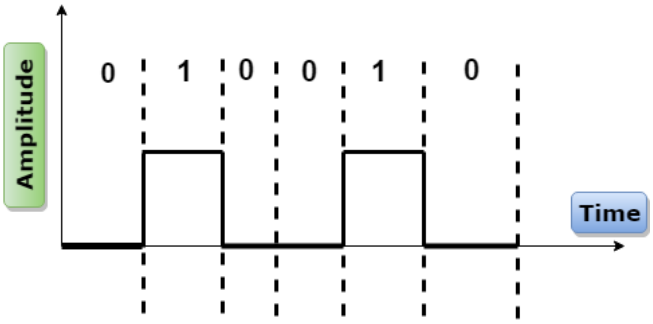


The various types of Digital-to-digital encoding schemes are shown in the following figure.



Unipolar: This type of encoding is known as unipolar encoding as it uses only one polarity.

In Unipolar encoding, the polarity is assigned to the 1 binary state. In this, 1s are represented as a positive value and 0s are represented as a zero value. In other words, '1' is considered as a high voltage and '0' is considered as a zero voltage. Unipolar encoding is simpler and inexpensive to implement.



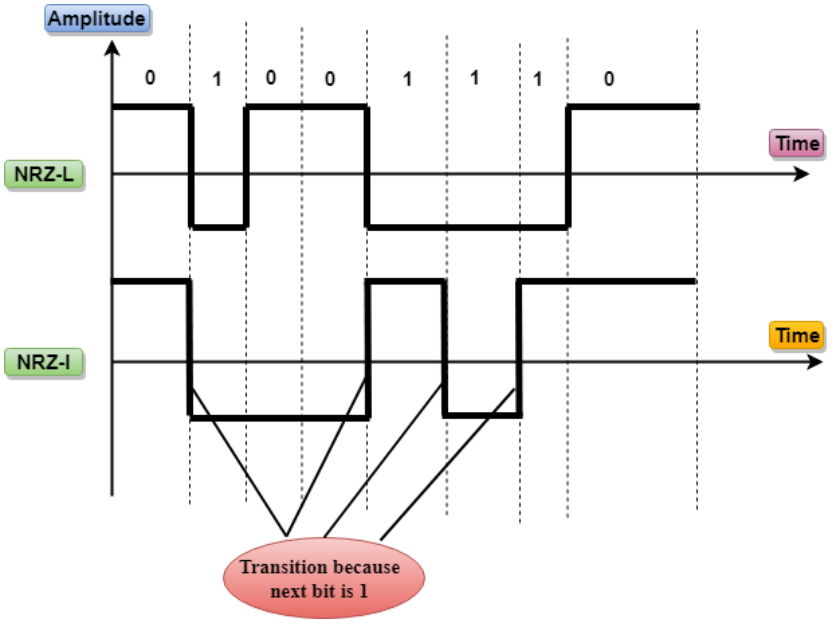
Unipolar encoding has two problems that make this scheme less desirable:

- DC Component
- Synchronization

Polar: Polar encoding is an encoding scheme that uses two voltage levels: one is positive, and another is negative. By using two voltage levels, an average voltage level is reduced, and the DC component problem of unipolar encoding scheme is reduced. Polar encoding scheme is of three types: NRZ, RZ and Biphase.

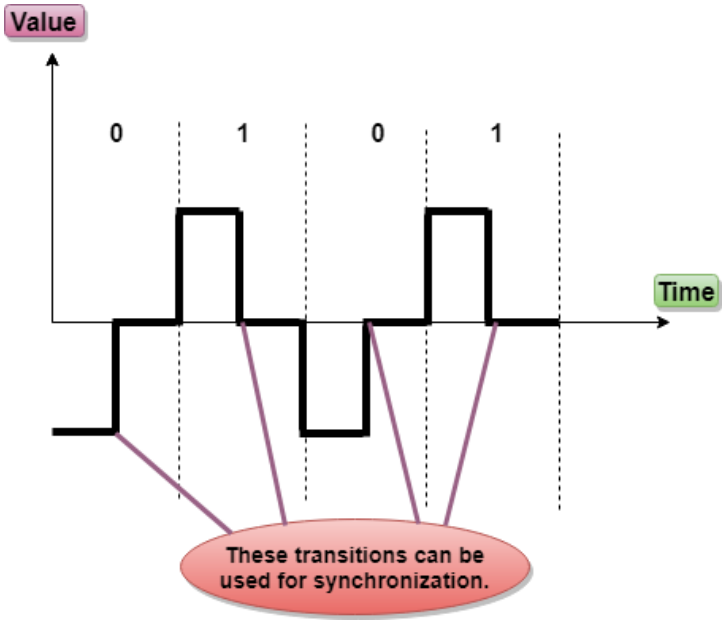
NRZ: NRZ stands for Non-return zero. In NRZ encoding, the level of the signal can be represented either positive or negative. The two most common methods used in NRZ are NRZ-L and NRZ-I.

NRZ-L: In NRZ-L encoding, the level of the signal depends on the type of the bit that it represents. If a bit is 0 or 1, then their voltages will be positive and negative respectively. Therefore, we can say that the level of the signal is dependent on the state of the bit.



NRZ-I: NRZ-I is an inversion of the voltage level that represents 1 bit. Here, a transition occurs between the positive and negative voltage if the next bit is 1. In this scheme, 0 bit represents no change and 1 bit **represents a change in voltage level.**

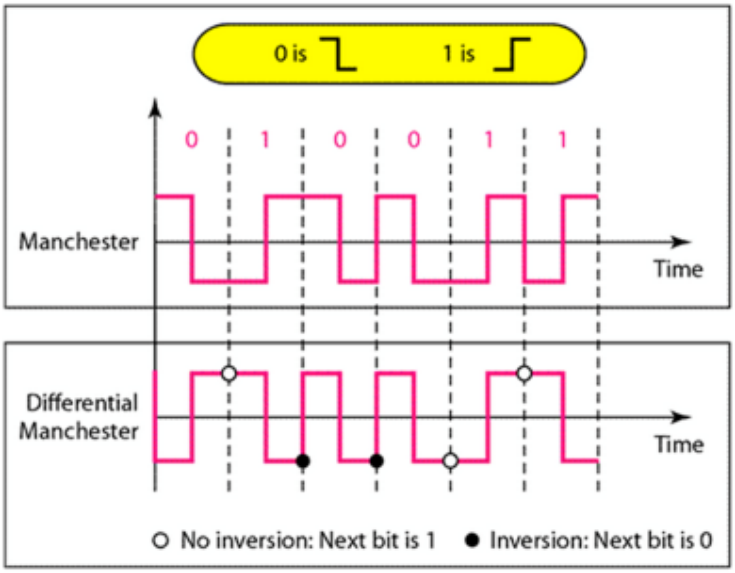
RZ: RZ stands for Return to zero. There must be a signal change for each bit to achieve synchronization. It is an encoding scheme that provides three values, positive voltage represents 1, the negative voltage represents 0, and zero voltage represents none. Here, in the middle of each bit, the signal returns to zero. 1 bit is represented by positive-to-zero and 0 bit is represented by negative-to-zero.



Disadvantage of RZ: It performs two signal changes to encode one bit that acquires more bandwidth.

Biphase: Biphase is an encoding scheme in which signal changes at the middle of the bit interval but does not return to zero. Biphase encoding is implemented in two different ways; Manchester and Differential Manchester

Manchester: It changes the signal at the middle of the bit interval but does not return to zero for synchronization. Negative-to-positive transition represents binary 1, and positive-to-negative transition represents 0. Manchester has the same level of synchronization as RZ scheme except that it has two levels of amplitude.

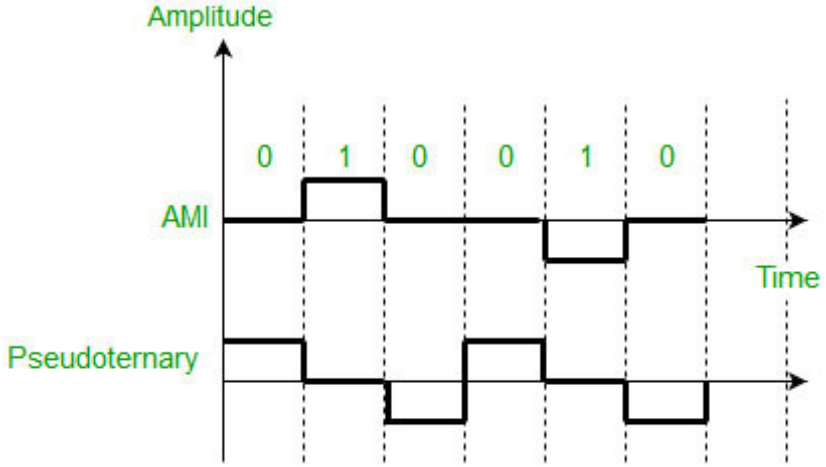


Differential Manchester: It changes the signal at the middle of the bit interval for synchronization. But the presence or absence of the transition at the beginning of the interval determines the bit. A transition means binary 0 and no transition means binary 1.

Bipolar: Bipolar encoding scheme represents three voltage levels; positive, negative and zero. The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative. Bipolar can be classified as Alternate Mark Inversion (AMI) and Pseudoternary.

Alternate Mark Inversion (AMI) – A neutral zero voltage represents binary 0. Binary 1's are represented by alternating positive and negative voltages.

Pseudoternary – Bit 1 is encoded as a zero voltage and the bit 0 is encoded as alternating positive and negative voltages i.e., opposite of AMI scheme.



<u>Analog-to-Digital Conversion</u>: When an analog signal is digitalized, this is called an analog-to-digital conversion. Suppose human voice is sent in the form of an analog signal, we need to digitalize the analog signal which is less prone to noise. In analog-to-digital conversion, the information contained in a continuous wave form is converted in digital pulses. The techniques for **analog signal to digital data** are:

- Pusle Code Modulation
- Delta Modulation
- Adaptive Delta Modulation

Pulse Code Modulation: The most common technique to change an analog signal to digital data is called pulse code modulation (PCM). A PCM encoder has the following three processes:

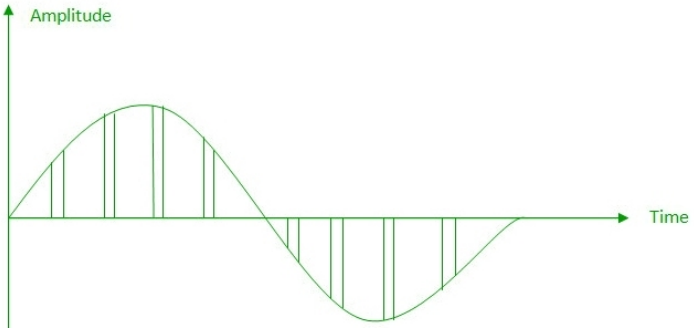
- Sampling
- Quantization
- Encoding

Sampling – The first step in PCM is sampling. Sampling is a process of measuring the amplitude of a continuous-time signal at discrete instants, converting the continuous signal into a discrete signal. The sampling process is sometimes referred to as **pulse amplitude modulation (PAM).** There are three sampling methods:

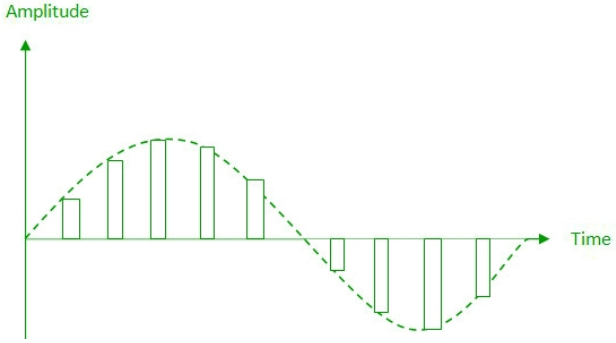
(i) Ideal Sampling: This is an ideal sampling method and cannot be easily implemented.

(ii) Natural Sampling: Natural Sampling is a practical method of sampling in which pulse have finite width equal to T.The result is a sequence of samples that retain the shape of the analog signal.

(iii) Flat top sampling: In comparison to natural sampling flat top sampling can be easily obtained. In this sampling technique, the top of the samples remains constant by using a circuit. This is the most common sampling method used.



(Natural Sampling)



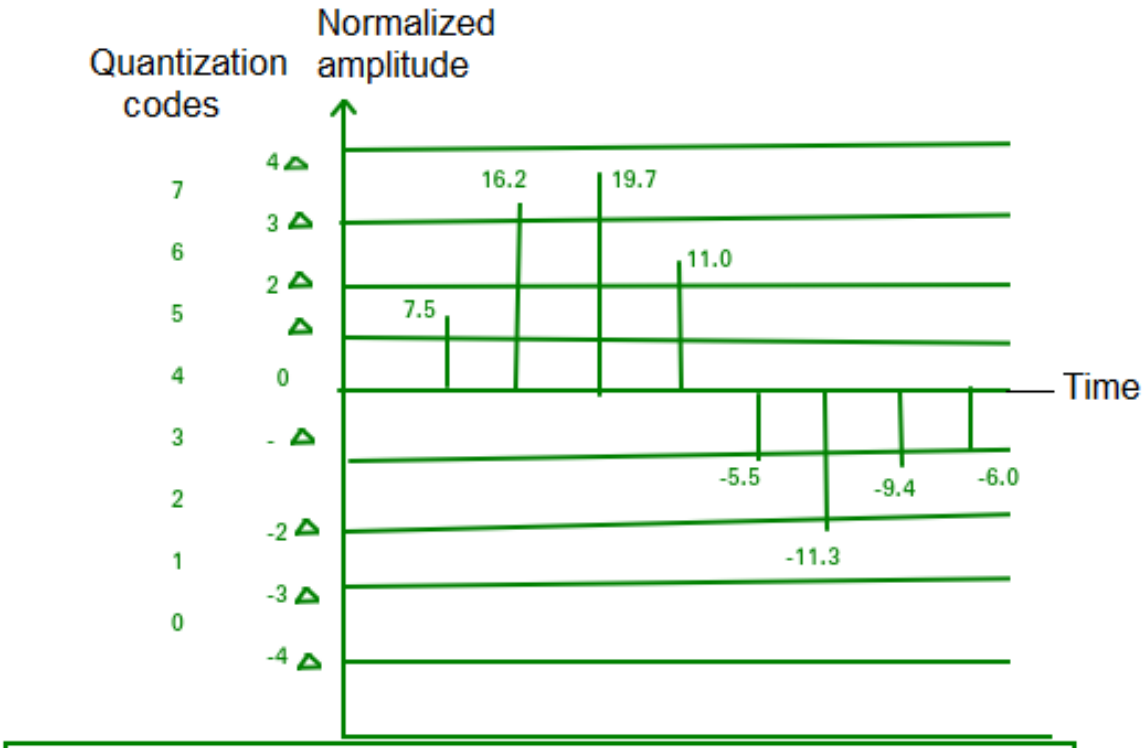
(Flat top sampling)

Nyquist Theorem: One important consideration is the sampling rate or frequency. According to Nyquist theorem, the **minimum sampling rate must be at least twice the highest frequency** contained in the signal.

Quantization: – The result of sampling is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal. Following are the steps in Quantization:

1. We assume that the signal has amplitudes between Vmax and Vmin
2. We divide the range into L zones, each of height d where, Δ= (Vmax- Vmin)/ L
3. We assign quantized values of 0 to L–1 t the midpoint of each zone
4. We approximate the value of the sample amplitude to the quantized values.

Exapmle: Assume that we have a smapled signal and the sample amplitudes are between -20 and +20V. We decide to have eight levels (L=8). Thsi means that Δ=40/8 = 5V. This is shown in the following example



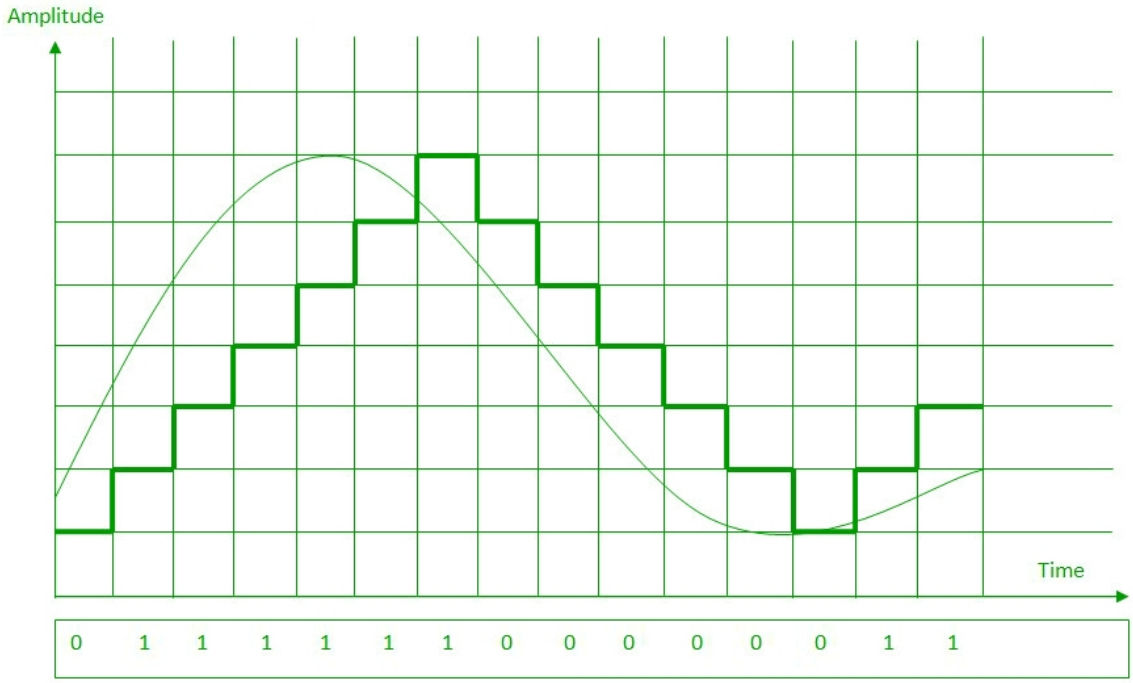
Normalised PAM value :	1.50	3.24	3.94	2.20	-1.10	2.26	-1.88	-1.20
Quantized Value :	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized Error :	0	0.26	-0.44	0.30	-0.40	-0.24	0.38	-0.30
Quantization Code :	5	7	7	6	2	1	2	2
Encoded Words :	101	111	111	110	010	001	010	010

Encoding: – The digitization of the analog signal is done by the encoder. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n bit code. Encoding also minimizes the bandwidth used.

Delta Modulation: Since PCM is a very complex technique, other techniques have been developed to reduce the complexity of PCM. The simplest is delta Modulation. Delta Modulation finds the change from the previous value.

Modulator – The modulator is used at the sender site to create a stream of bits from an analog signal. The process records a small positive change called delta. If the delta is positive, the process records a 1 else the process records a 0. The modulator builds a second signal that resembles a staircase. The input signal is then compared with this gradually made staircase signal. We have the following rules for output:

1. If the input analog signal is higher than the last value of the staircase signal, increase delta by 1, and the bit in the digital data is 1.
2. If the input analog signal is lower than the last value of the staircase signal, decrease delta by 1, and the bit in the digital data is 0.

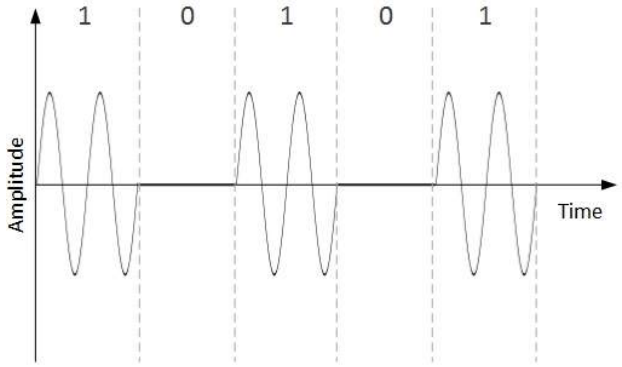


Adaptive Delta Modulation: The performance of a delta modulator can be improved significantly by making the step size of the modulator assume a time-varying form. A larger step-size is needed where the message has a steep slope of modulating signal and a smaller step-size is needed where the message has a small slope. The size is adapted according to the level of the input signal. This method is known as adaptive delta modulation (ADM).

Digital-to-Analog Conversion: When data from one computer is sent to another via some analog carrier, it is first converted into analog signals. Analog signals are modified to reflect digital data. An analog signal is characterized by its amplitude, frequency, and phase. There are three kinds of digital-to-analog conversions:

- Amplitude Shift Keying (ASK)
- Frequency Shift Keying (FSK)
- Phase Shift Keying (PSK)
- Quadrature Amplitude Modulation (ASK+PSK)

Amplitude Shift Keying (ASK): In this conversion technique, the amplitude of analog carrier signal is modified to reflect binary data. When binary data represents digit 1, the amplitude is held; otherwise it is set to 0. Both frequency and phase remain same as in the original carrier signal.



Advantages of amplitude shift Keying –

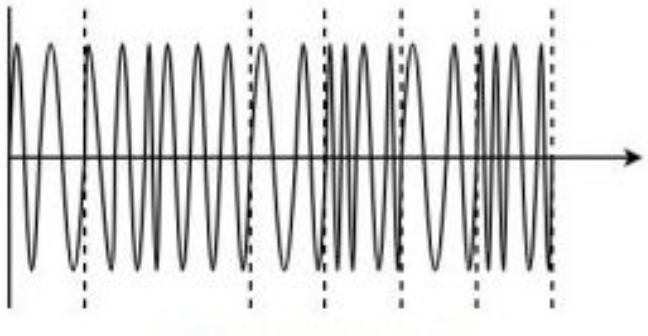
- It can be used to transmit digital data over optical fiber.

- The receiver and transmitter have a simple design which also makes it comparatively inexpensive.
- It uses lesser bandwidth as compared to FSK thus it offers high bandwidth efficiency.

Disadvantages of amplitude shift Keying –

- It is susceptible to noise interference and entire transmissions could be lost due to this.
- It has lower power efficiency.

Frequency Shift Keying (FSK): In this conversion technique, the frequency of the analog carrier signal is modified to reflect binary data. This technique uses two frequencies, f1 and f2. One of them, for example f1, is chosen to represent binary digit 1 and the other one is used to represent binary digit 0.



Both amplitude and phase of the carrier wave are kept intact.

Advantages of frequency shift Keying –

- Frequency shift keying modulated signal can help avoid the noise problems beset by ASK.
- It has lower chances of an error.
- It provides high signal to noise ratio.
- The transmitter and receiver implementations are simple for low data rate application.

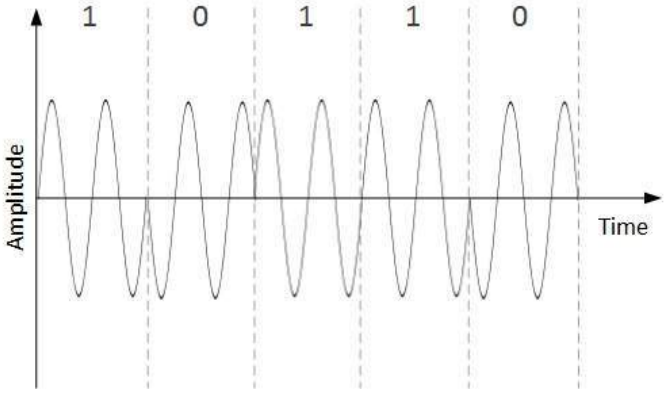
Disadvantages of frequency shift Keying –

- It uses larger bandwidth as compared to ASK thus it offers less bandwidth efficiency.
- It has lower power efficiency.

Phase Shift Keying (PSK): In this modulation the phase of the analog carrier signal is modified to reflect binary data.The amplitude and frequency of the carrier signal remains constant.It is further categorized as follows.

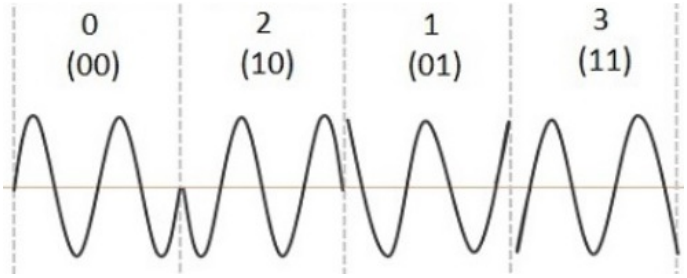
- Binary Phase Shift Keying (BPSK)
- Quadrature phase shift keying (QPSK)

Binary Phase Shift Keying (BPSK): BPSK also known as phase reversal keying or 2PSK is the simplest form of phase shift keying. The Phase of the carrier wave is changed according to the two binary inputs. In Binary Phase shift keying, difference of 180 phase shift is used between binary 1 and binary 0.



This is regarded as the most robust digital modulation technique and is used for long distance wireless communication.

Quadrature phase shift keying (QPSK): This technique is used to increase the bit rate i.e we can code two bits onto one single element. It uses four phases to encode two bits per symbol. QPSK uses phase shifts of multiples of 90 degrees.



It has double data rate carrying capacity compare to BPSK as two bits are mapped on each constellation points.

Advantages of phase shift Keying –

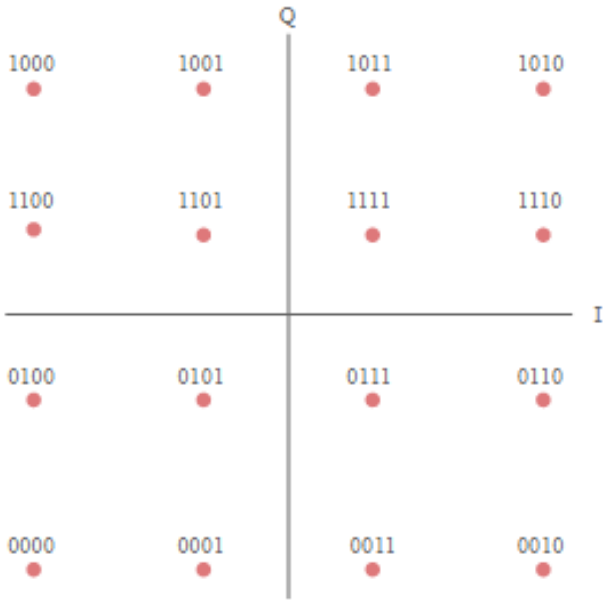
- It is a more power efficient modulation technique as compared to ASK and FSK.
- It has lower chances of an error.
- It allows data to be carried along a communication signal much more efficiently as compared to FSK.

Disadvantages of phase shift Keying –

- It offers low bandwidth efficiency.
- The detection and recovery algorithm of binary data is very complex.
- It is a non coherent reference signal.

Quadrature Amplitude Modulation (ASK+PSK): *Quadrature Amplitude Modulation combines amplitude & phase changes to give additional capacity & is widely used for data communications.* Basic signals exhibit only two positions which allow the transfer of either a 0 or 1. Using QAM there are many different points that can be used, each having defined values of phase and amplitude. This is known as a **constellation diagram**. The different positions are assigned different values, and in this way a single signal is able to transfer data at a much higher rate.

To provide an example of how QAM operates, the constellation diagram shown here indicates the values associated with the different states for a **16QAM (4 Phase & 4 Amplitude) signal**. From this it can be seen that a continuous bit stream may be grouped into fours and represented as a sequence. Normally the lowest order QAM encountered is 16QAM. The reason for this being the lowest order normally encountered is that 2QAM is the same as binary phase-shift keying (BPSK) and 4QAM is the same as quadrature phase-shift keying (QPSK).



Advantages of QAM:

- Increase the efficiency of transmission.

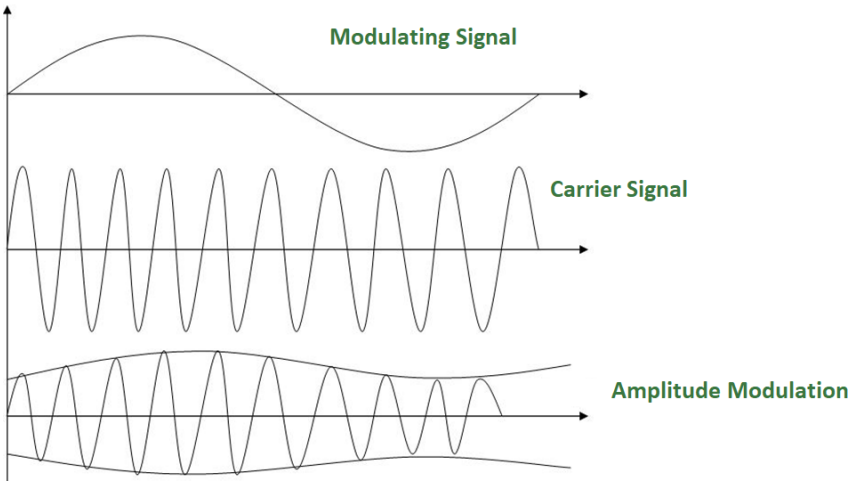
Disadvantages of QAM:

- It is more susceptible to noise because the states are closer together

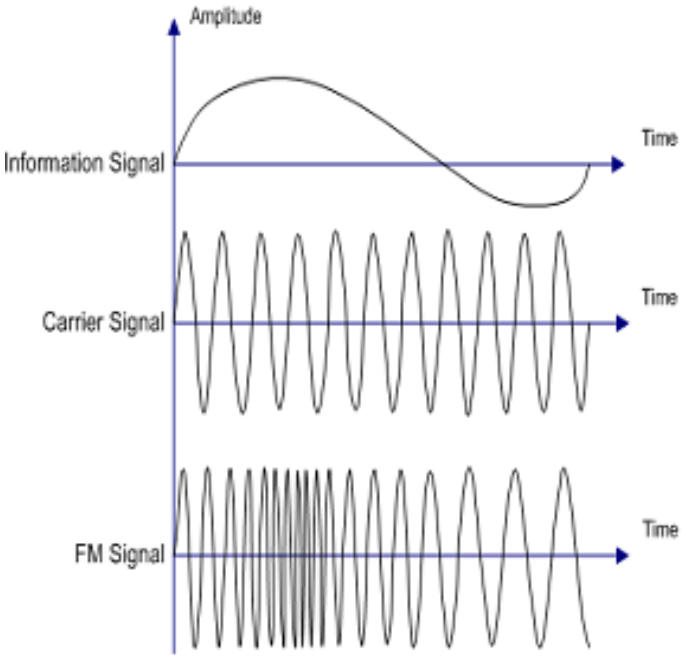
Analog-to-Analog Conversion: Analog-to-analog conversion, or modulation, is the representation of analog information by an analog signal. It is a process by virtue of which a characteristic of carrier wave is varied according to the instantaneous amplitude of the modulating signal. This modulation is generally needed when a **bandpass channel** is required. Bandpass is a range of frequencies which are transmitted through a bandpass filter that allows specific frequencies to pass and prevents signals of unwanted frequencies. Analog to Analog conversion can be done in three ways:

1. Amplitude Modulation (AM)
2. Frequency Modulation (FM)
3. Phase Modulation (PM)

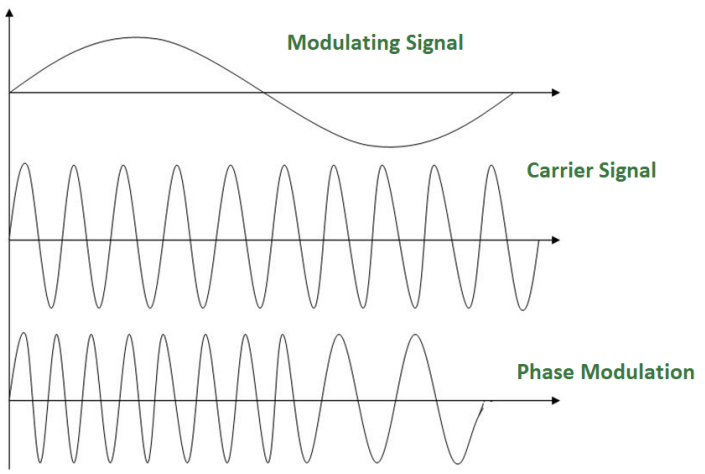
1. Amplitude Modulation: The modulation in which the amplitude of the carrier wave is varied according to the instantaneous amplitude of the modulating signal keeping phase and frequency as constant. The figure shows the concept of amplitude modulation.



2. Frequency Modulation: The modulation in which the frequency of the carrier wave is varied according to the instantaneous amplitude of the modulating signal keeping phase and amplitude as constant. The figure shows the concept of frequency modulation. FM is normally implemented by using a voltage-controlled oscillator (VCO) as with FSK. The frequency of the oscillator changes according to the input voltage which is the amplitude of the modulating signal.



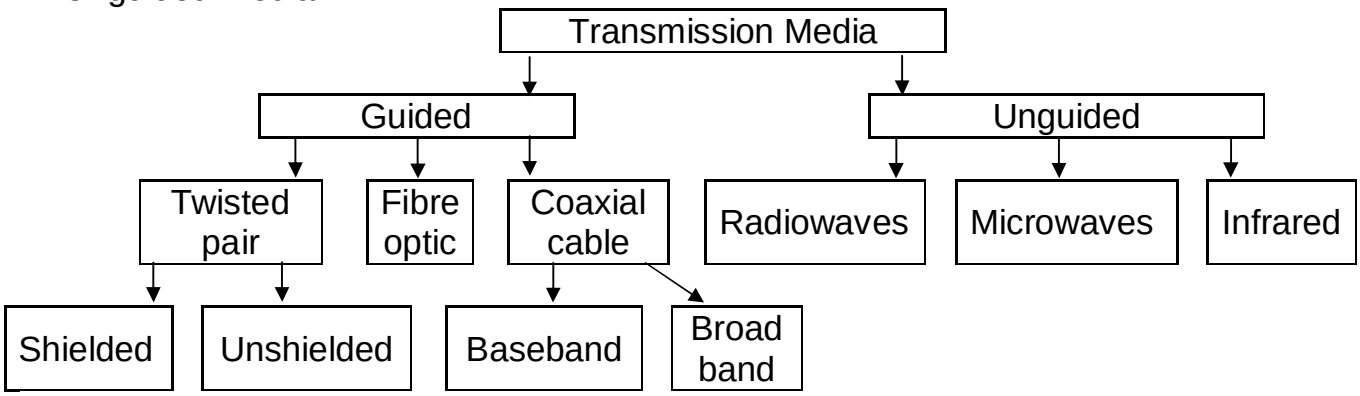
3. PHASE MODULATION: The modulation in which the phase of the carrier wave is varied according to the instantaneous amplitude of the modulating signal keeping amplitude and frequency as constant. The figure shows the concept of phase modulation. Phase modulation is practically similar to Frequency Modulation, but in Phase modulation frequency of the carrier signal is not increased.



It is normally implemented by using a voltage-controlled oscillator along with a derivative. The frequency of the oscillator changes according to the **derivative of the input voltage which is the amplitude** of the modulating signal.

<u>Transmission media</u>: A transmission medium is a physical path between the transmitter and the receiver i.e it is the channel through which data is sent from one place to another. Transmission Media is broadly classified into two types.

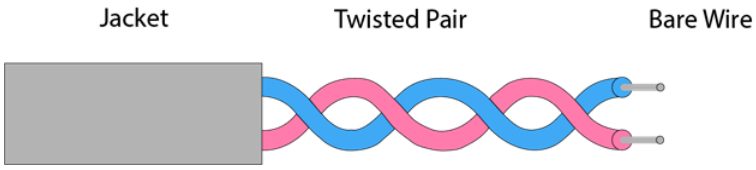
1. Guided Media
2. Unguided media



1. Guided Media: It is also referred to as Wired or Bounded transmission media. There are three types of guided media.

a) Twisted pair
b) Coaxial cable
c) Fibre optic

a) Twisted Pair: It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath.



They are the most widely used Transmission Media. Twisted Pair is of two types.

i. Unshielded Twisted Pair (UTP)
ii. Shielded Twisted Pair (STP):

i. Unshielded Twisted Pair (UTP): This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Advantages:

- Least expensive
- Easy to install
- High-speed capacity

Disadvantage:

- Susceptible to external interference
- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation

ii. Shielded Twisted Pair (STP): This type of cable consists of a special jacket to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages:

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

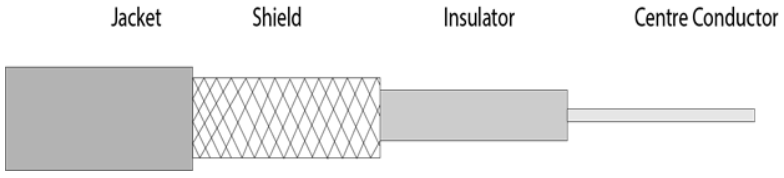
Disadvantage:

- Comparatively difficult to install and manufacture
- More expensive
- Bulky

b) Coaxial Cable – Coaxial cable consists of three layers of elements; core, cladding and jacket.

- **Core:** The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into the fibre.
- **Cladding:** The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre.
- **Jacket:** The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.

The coaxial cable transmits information in two modes: Baseband mode (dedicated cable bandwidth) and Broadband mode (cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.



Advantages:

- High Bandwidth
- Better noise Immunity
- Easy to install and expand
- Inexpensive

Disadvantages:

- Single cable failure can disrupt the entire network

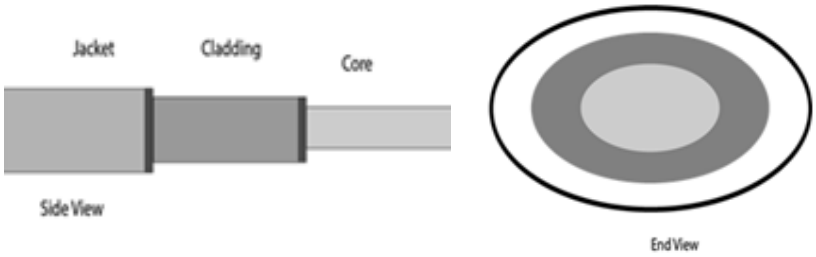
c) Fibre Optic Cable – It uses the concept of reflection of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for the transmission of large volumes of data. The cable can be unidirectional or bidirectional. The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.

Advantages:

- Increased capacity and bandwidth
- Lightweight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages:

- Difficult to install and maintain
- High cost
- Fragile



2. Unguided Media: It is also referred to as Wireless or unbounded transmission media. No physical medium is required for the transmission of electromagnetic signals. Unguided media has the following features.

- The signal is broadcasted through air
- Less Secure
- Used for larger distances

There are 3 types of Signals transmitted through unguided media:

(i) Radiowaves – These are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range: 3 KHz – 1GHz. AM and FM radios and cordless phones use Radiowaves for transmission.
It is further Categorized as (i) Terrestrial and (ii) Satellite.

(ii) Microwaves – It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz – 300GHz. These are majorly used for mobile phone communication and television distribution.

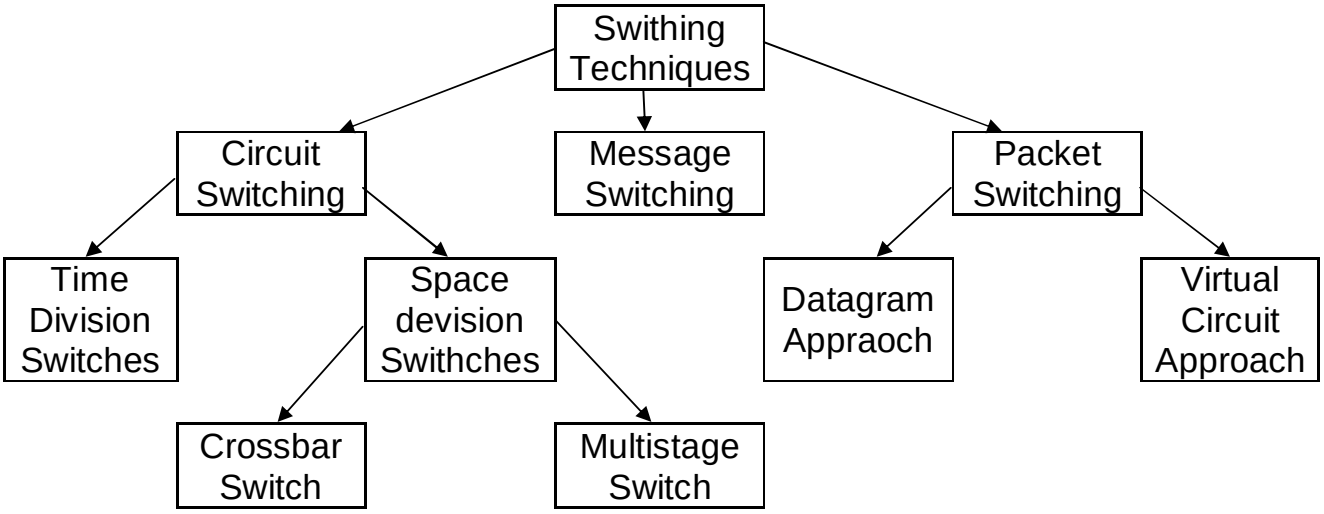
(iii) Infrared – Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems. Frequency Range: 300GHz – 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.

Switching Techniques: Switched communication networks are those in which data transferred from source to destination is routed between various intermediate nodes. Switching is the technique by which nodes control or switch data to transmit it between specific points on a network.

Why is Switching Concept required? Switching concept is developed because of the following reasons:

- It increases bandwidth and overall performance of the network by reducing the traffic on the network.
- To overcome collision of packets when more than one device transmits the messge over the same physical media.
- In large networks, there can be multiple paths from sender to receiver. Switching technique is used to connect the systems for making one-to-one communication.

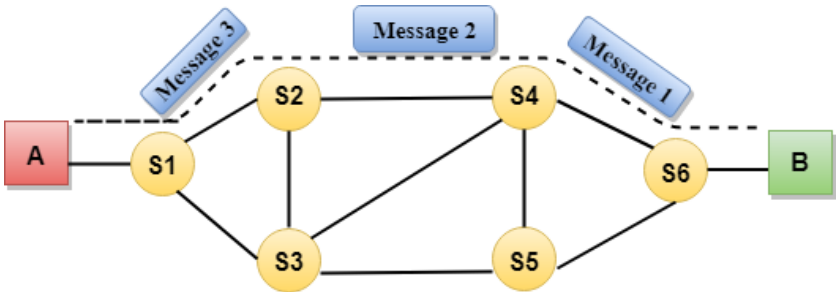
Classification of Switching Techniques:



Circuit Switching- Circuit switching is a switching technique that **establishes a dedicated path between sender and receiver**. In Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated. When any user wants to send the data, voice, video, a request signal is sent to the receiver then the receiver sends back the acknowledgment to ensure the availability of the dedicated path. After receiving the acknowledgment, data is transferred through the dedicated path. It is used in public telephone network. It is used for voice transmission. Communication through circuit switching has 3 phases:

- Circuit establishment
- Data transfer
- Circuit Disconnect

Circuit Switching can use either of the two technologies:



1. Space Division Switches
2. Time Division Switches

Space Division Switches: Space Division Switching is a circuit switching technology in which a single transmission path is accomplished in a switch by using a **physically separate set of crosspoints.** Space Division Switching can be achieved by metallic crosspoint or semiconductor gate that can be enabled or disabled by a control unit. Space Division Switching has high speed, high capacity, and nonblocking switches. Space Division Switches can be categorized in two ways:

- Crossbar Switch
- Multistage Switch

Crossbar Switch: The Crossbar switch is a switch that has n input lines and n output lines. The crossbar switch has n^2 intersection points known as **crosspoints.**

Disadvantage of Crossbar switch: The number of crosspoints increases as the number of stations is increased. Therefore, it becomes very expensive for a large switch. The solution to this is to use a multistage switch.

Multistage Switch: Multistage Switch is made by splitting the crossbar switch into the smaller units and then interconnecting them. It reduces the number of crosspoints. If one path fails, then there will be an availability of another path.

Time Division Switches: Time division switching comes under digital switching techniques, where the incoming and outgoing signals when received and re-transmitted in a different time slot. The main difference between space division multiplexing and time division multiplexing is **sharing of Crosspoints**. Crosspoints are not shared in space division switching, whereas they can be shared in time division multiplexing, for shorter periods. This helps in reassigning the Crosspoints and its associated circuitry for other connections as well.

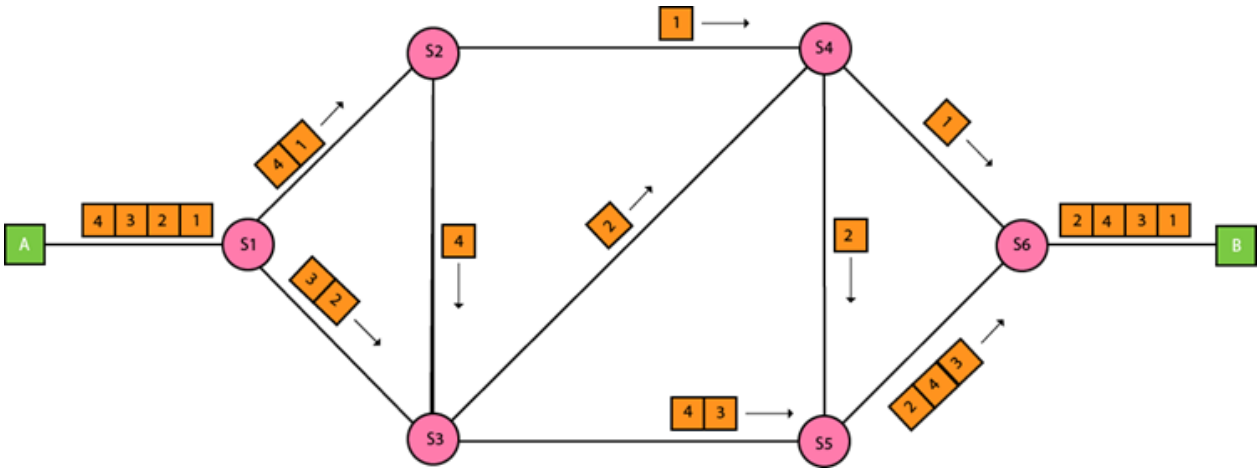
Advantages of Circuit Switching:

- It is suitable for long continuous transmission, since a continuous transmission route is established, that remains throughout the conversation.
- The dedicated path ensures a steady data rate of communication.
- No intermediate delays are found once the circuit is established. So, they are suitable for real time communication of both voice and data transmission.

Disadvantages of Circuit Switching:

- Time required to establish connection may be high.
- It is more expensive than other switching techniques as a dedicated path is required for each connection.
- There is underutilization of system resources. Once resources are allocated to a particular connection, they cannot be used for other connections.

Packet Switching: The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces known as packets, and they are sent individually. The packets are given a unique number to identify their order at the receiving end. Every packet contains some information in its headers such as source address, destination address and sequence number.



There are two approaches to Packet Switching-

- Datagram Packet switching
- Virtual Circuit Switching

Datagram Packet switching:

- It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity. Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.
- In Datagram Packet Switching technique, the **path is not fixed**.
- Intermediate nodes take the routing decisions to forward the packets.
- Datagram Packet Switching is also known as **connectionless switching**.

Virtual Circuit Switching

- Virtual Circuit Switching is also known as **connection-oriented switching**.
- In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.
- Call request and call accept packets are used to establish the connection between sender and receiver.
- In this case, the **path is fixed** for the duration of a **logical connection**.

Differences between Datagram approach and Virtual Circuit approach

Datagram approach	Virtual Circuit approach
Node takes routing decisions to forward the packets.	Node does not take any routing decision.
Congestion cannot occur as all the packets travel in different directions.	Congestion can occur when the node is busy, and it does not allow other packets to pass through.
It is more flexible as all the packets are treated as an independent entity.	It is not very flexible.

Advantages of Packet Switching:

- Delay in delivery of packets is less, since packets are sent as soon as they are available.
- Switching devices don't require massive storage, since they don't have to store the entire messages before forwarding them to the next node.
- Data delivery can continue even if some parts of the network face link failure. Packets can be routed via other paths.
- It allows simultaneous usage of the same channel by multiple users.
- It ensures better bandwidth usage as a number of packets from multiple sources can be transferred via the same link.

Disadvantages of Packet Switching:

- They are unsuitable for applications that cannot afford delays in communication like high quality voice calls.
- **Packet switching** high installation costs.
- They require complex protocols for delivery.
- Network problems may introduce errors in **packets**, delay in delivery of **packets** or loss of **packets**.

Difference between Circuit Switching and Packet Switching

Circuit Switching	Packet Switching
In circuit switching there are 3 phases: i) Connection Establishment. ii) Data Transfer. iii) Connection Released.	In Packet switching directly data transfer takes place.
In circuit switching, each data unit know the entire path address which is provided by the source.	In Packet switching, each data unit just know the final destination address intermediate path is decided by the routers.
Delay between data units in circuit switching is uniform.	Delay between data units in packet switching is not uniform.
Resource reservation is the feature of circuit switching because path is fixed for data transmission.	There is no resource reservation because bandwidth is shared among users.
Circuit switching is more reliable.	Packet switching is less reliable.
Wastage of resources are more in Circuit Switching	Less wastage of resources as compared to Circuit Switching
It is not a store and forward technique.	It is a store and forward technique.
Transmission of the data is done by the source.	Transmission of the data is done not only by the source, but also by the intermediate routers.
Congestion can occur during connection establishment time.	Congestion can occur during data transfer phase.